

31. MAC Control

31.1 Overview

This clause specifies an optional MAC Control sublayer (MAC Control) for use with the CSMA/CD MAC. MAC Control provides for real-time control and manipulation of MAC sublayer operation. This clause specifies a generalized architecture and protocol for MAC Control. Specific implementations of control functions using this protocol are specified in the normative annexes to this clause. The MAC Control protocol is specified such that it can support new functions to be implemented and added to this standard in the future.

Non-realtime, or quasistatic control (e.g., configuration of MAC operational parameters) is provided by Layer Management.

Operation of the MAC Control sublayer is transparent to the CSMA/CD MAC.

The body of this clause and its associated annexes contain state diagrams, including definitions of variables, constants, and functions. Should there be a discrepancy between a state diagram and descriptive text, the state diagram shall prevail. The notation used in the state diagrams follows the conventions of 21.5.

31.2 Layer architecture

The MAC Control sublayer is a client of the CSMA/CD MAC. Figure 31–1 depicts the architectural positioning of the MAC Control sublayer with respect to the CSMA/CD MAC and the MAC Control client. MAC Control clients may include the Bridge Relay Entity, LLC, or other applications.

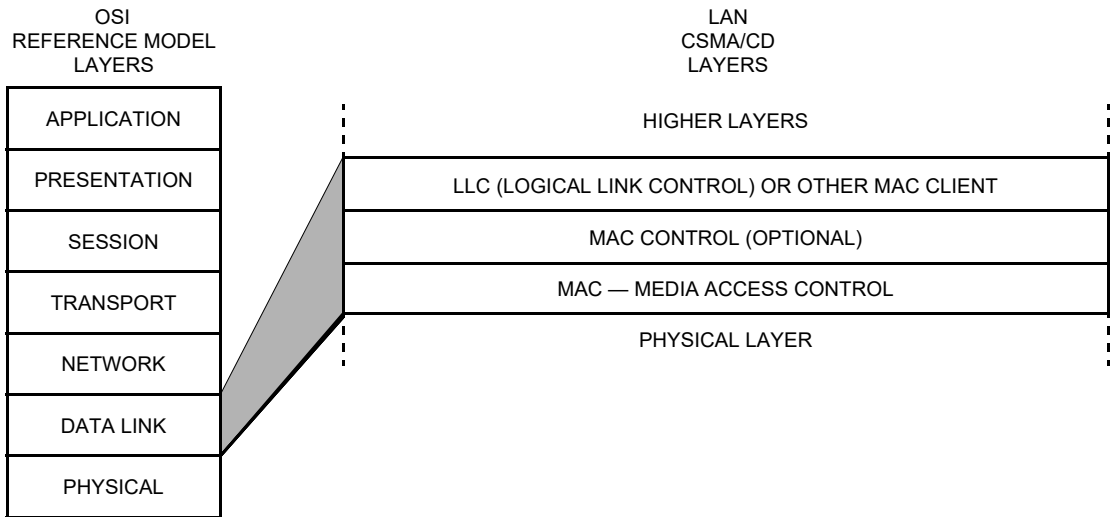


Figure 31–1—Architectural positioning of MAC Control sublayer

31.3 Support by interlayer interfaces

This subclause describes how the MAC Control sublayer uses the MAC service interface specified in Clause 2. The optional MAC Control sublayer is inserted between the MAC sublayer and its MAC client. The MAC Control sublayer uses the MAC service interface to interface to the MAC client and to the MAC.

Figure 31–2 depicts the usage of interlayer interfaces by the MAC Control sublayer. Devices that implement the MAC Control sublayer shall support the MAC service primitives, MA_CONTROL.request and MA_CONTROL.indication, as specified in this clause.

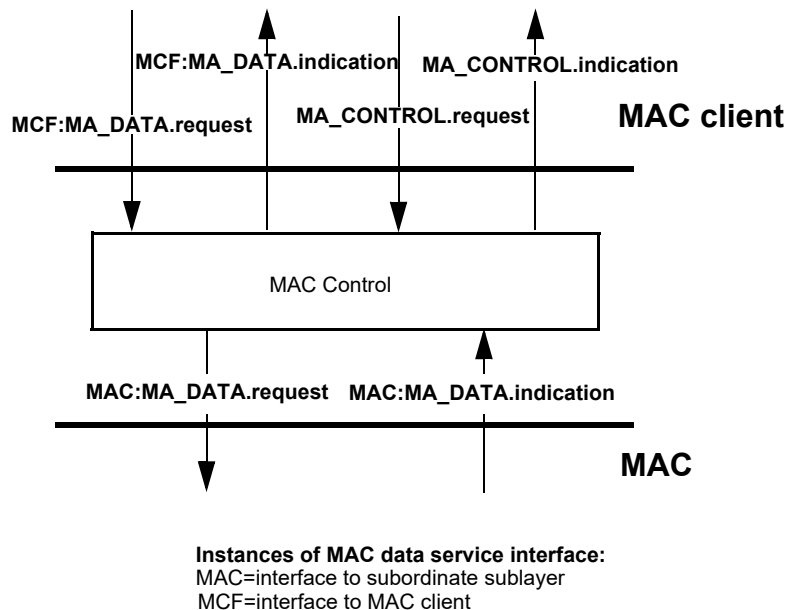


Figure 31–2—MAC Control sublayer support of interlayer service interfaces

Clients of the MAC Control sublayer may generate either MCF:MA_CONTROL.request or MCF:MA_DATA.request primitives. MA_CONTROL.request primitives generated by MAC clients are interpreted by the MAC Control sublayer, and may result in the generation of MAC:MA_DATA.request calls to the MAC sublayer, or other actions as necessary to support the requested MAC Control sublayer function. Based upon the state of the MAC Control sublayer, MCF:MA_DATA.request primitives may cause the immediate generation of a MAC:MA_DATA.request call to the MAC sublayer, or be delayed, discarded, or modified in order to perform the requested MAC Control function.

All MAC frames validly received by the CSMA/CD MAC are passed to the MAC Control sublayer for interpretation. If the MAC frame is destined for the MAC client, the MAC Control sublayer generates an MCF:MA_DATA.indication primitive, providing complete transparency for normal data exchange between MAC clients. If the MAC frame is destined for the MAC Control sublayer entity, it is interpreted and acted on internal to the MAC Control sublayer. This may result in state changes within the MAC Control sublayer, the generation of MA_CONTROL.indication primitives, or other actions as necessary to support the MAC Control sublayer function. MAC Control sublayer functions shall always sink MAC Control frames.

In the MAC:MA_DATA.indication primitive, MAC frames destined for the MAC Control sublayer (MAC Control frames) are distinguished from MAC frames destined for MAC clients by a unique Length/Type field identifier.

The MAC Control sublayer generates MA_CONTROL.indication primitives to its client, signaling the current value of internal state variables. Each MAC Control function implemented may have its own function-specific state indications.

31.3.1 MA_CONTROL.request

Implementation of the MA_CONTROL.request primitive is mandatory.

31.3.1.1 Function

This primitive defines the transfer of control commands from a MAC client entity to the local MAC Control sublayer entity.

31.3.1.2 Semantics of the service primitive

The semantics of the primitive are as follows:

```
MA_CONTROL.request    (
                        destination_address,
                        opcode,
                        request_operand_list
                        )
```

The destination_address parameter may specify either an individual or a group MAC entity address. It has to contain sufficient information to create the DA field that is prepended to the MAC frame by the local MAC sublayer entity. The opcode specifies the control operation requested by the MAC client entity. The request_operand_list is an opcode-specific set of parameters.

The valid opcodes and their respective meanings are defined in Annex 31A.

31.3.1.3 When generated

This primitive is generated by a MAC client whenever it wishes to use the services of the optional MAC Control sublayer entity.

31.3.1.4 Effect of receipt

The effect of receipt of this primitive by the MAC Control sublayer is opcode-specific. (See Annex 31A.)

31.3.2 MA_CONTROL.indication

Implementation of the MA_CONTROL.indication primitive is mandatory.

31.3.2.1 Function

This primitive defines the transfer of control status indications from the MAC Control sublayer entity to the MAC client entity.

31.3.2.2 Semantics of the service primitive

The semantics of the primitive are as follows:

```
MA_CONTROL.indication (
                        opcode,
                        indication_operand_list
                        )
```

The elements of the indication_operand_list are opcode-specific, and specified in Annex 31A.

31.3.2.3 When generated

The MA_CONTROL.indication is generated by the MAC Control sublayer under conditions specific to each MAC Control operation.

31.3.2.4 Effect of receipt

The effect of receipt of this primitive by the MAC client is not specified in this clause. See the list of MAC control functions in Annex 31A.

31.4 MAC Control frames

MAC Control frames comprise MAC client data for the CSMA/CD MAC, as specified in Clause 3. They are encapsulated by the CSMA/CD MAC; that is, they are prepended by a Preamble and Start-of-Frame delimiter and appended by an FCS.

MAC Control frames are distinguished from other MAC frames only by their Length/Type field identifier.

31.4.1 MAC Control frame format

For any particular implementation of this standard, MAC Control frames are fixed length, containing minFrameSize–32 bits. The underlying MAC prepends the Preamble and Start-of-Frame delimiter fields, and appends the FCS. Figure 31–3 depicts the MAC Control frame format.

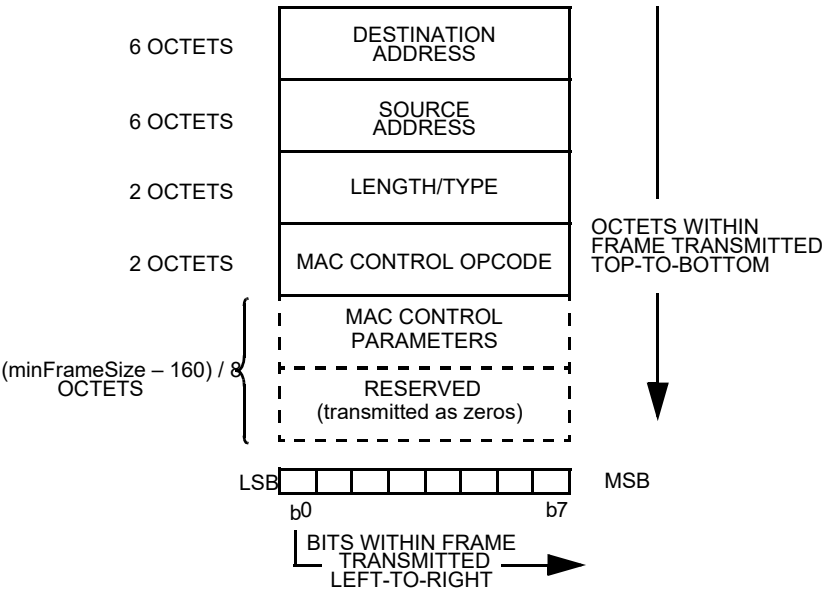


Figure 31–3—MAC Control frame format

31.4.1.1 Destination Address field

The Destination Address field of a MAC Control frame contains the 48-bit address of the station(s) for which the frame is intended. It may be an individual or multicast (including broadcast) address. Permitted values for the Destination Address field may be specified separately for each MAC Control opcode in the annexes to Clause 31.

31.4.1.2 Source Address field

The Source Address field of a MAC Control frame contains the 48-bit individual address of the station sending the frame.

31.4.1.3 Length/Type field

The Length/Type field of a MAC Control frame is a 2-octet field that shall contain the hexadecimal value: 88-08. This value carries the EtherType interpretation (see 3.2.6), and has been universally assigned for MAC Control of CSMA/CD LANs.

31.4.1.4 MAC Control Opcode field

The MAC Control Opcode field shall contain a 2-octet operation code indicating the MAC Control function. It defines the semantics of the MAC Control Parameters field specified in 31.4.1.5. Annex 31A contains the list of defined MAC Control opcodes and interpretations.

A MAC Control frame shall contain exactly one MAC Control opcode.

31.4.1.5 MAC Control Parameters field

The MAC Control Parameters field shall contain MAC Control opcode-specific parameters. This field may contain none, one, or more parameters as defined by the MAC Control Opcode. The opcode-specific semantics of the MAC Control Parameters field are defined in the normative annex specifying each MAC Control function.

The MAC Control Parameters field shall contain an integral number of octets. The length of the MAC Control Parameters field varies from a minimum of zero, to a maximum of $\text{minFrameSize} - 160$ bits. See 4.2.3.3 for a discussion of minFrameSize .

31.4.1.6 Reserved field

The Reserved field is used when the MAC Control parameters do not fill the fixed length MAC Control frame. The size of the Reserved field, if any, is determined by the size of the MAC Control Parameters field supplied by MAC Control and the minimum frame size parameter of the particular implementation.

The length of Reserved field required for a MAC Control Parameters field that is n octets long is $[\text{minFrameSize} - (8 \times n + 160)]$ bits. See 4.2.3.3 for a discussion of minFrameSize . The Reserved field is transmitted as all zeros.

31.5 Opcode-independent MAC Control sublayer operation

The MAC passes to the MAC Control sublayer all valid MAC frames via the `MA_DATA.indication` primitive. Invalid MAC frames are not passed to the MAC Control sublayer (see 3.4).

31.5.1 Frame parsing and data frame reception

Upon receipt, the MAC Control sublayer parses the incoming MAC frame to determine whether it is destined for the MAC client (data frame) or for a specific function within the MAC Control sublayer entity itself (MAC Control frame). MAC Control frames with a length of minFrameSize and a supported opcode field are interpreted and acted upon by the MAC Control sublayer.

A MAC frame that does not contain the unique Length/Type field specified in 31.4.1.3 is a data frame. The receipt of a data frame results in the generation of a MCF:MA_DATA.indication primitive by the MAC Control sublayer, with its parameters identical to the MAC:MA_DATA.indication primitive.

NOTE—For Length/Type field values in the range between maxBasicDataSize and minTypeValue, the behavior of the RemovePad function in the underlying MAC sublayer is unspecified. Frames with Length/Type field values in this range may or may not be passed up by the MAC sublayer.

MAC Control frames with a length greater than minFrameSize and a supported opcode field may be either discarded, or truncated to minFrameSize, interpreted, and acted upon. Unsupported MAC Control frames are discarded. Discarded frames are neither interpreted nor acted upon by the MAC Control sublayer.

31.5.2 Control frame reception

Validly received MAC Control frames are further parsed to determine the opcode. The location of the opcode within a valid MAC Control frame and its format are specified in 31.4.1.4 and Figure 31–3. If the MAC Control sublayer entity supports the function requested by the specified opcode, it interprets and acts upon the MAC Control frame in an opcode- and request_operand-specific manner. (See Annex 31A.) This action may change the state of the MAC Control sublayer, affecting its behavior with respect to data transmission requests by the MAC client, future control frame receptions, or control indications to the MAC client.

If the MAC Control sublayer entity does not support the function requested by the specified opcode, it discards the MAC Control frame. The discard of a frame in this manner may be reported to network management.

31.5.3 Opcode-independent MAC Control receive state diagram

The MAC Control sublayer shall implement the Receive state diagram specified in this subclause.

31.5.3.1 Constants

802.3_MAC_Control

The 16-bit Length/Type field value used for CSMA/CD MAC Control, specified in 31.4.1.3.

31.5.3.2 Variables

receiveEnabled

A Boolean set by Network Management to indicate that the station is permitted to receive from the network.

Values: true; Receiver is enabled by management
false; Receiver is disabled by management

DA

The destination address field parsed from the received frame.

SA

The source address field parsed from the received frame.

lengthOrType

The lengthOrType field parsed from the received frame.

data

The data payload field parsed from the received frame.

fcs

The fcs field parsed from the received frame.

fcsPresent

A Boolean set by the MAC sublayer. See 4.3.2.

mac_service_data_unit

The concatenation of lengthOrType, data.

ReceiveStatus

Indicates the status of the received frame. See 4.3.2.

opcode

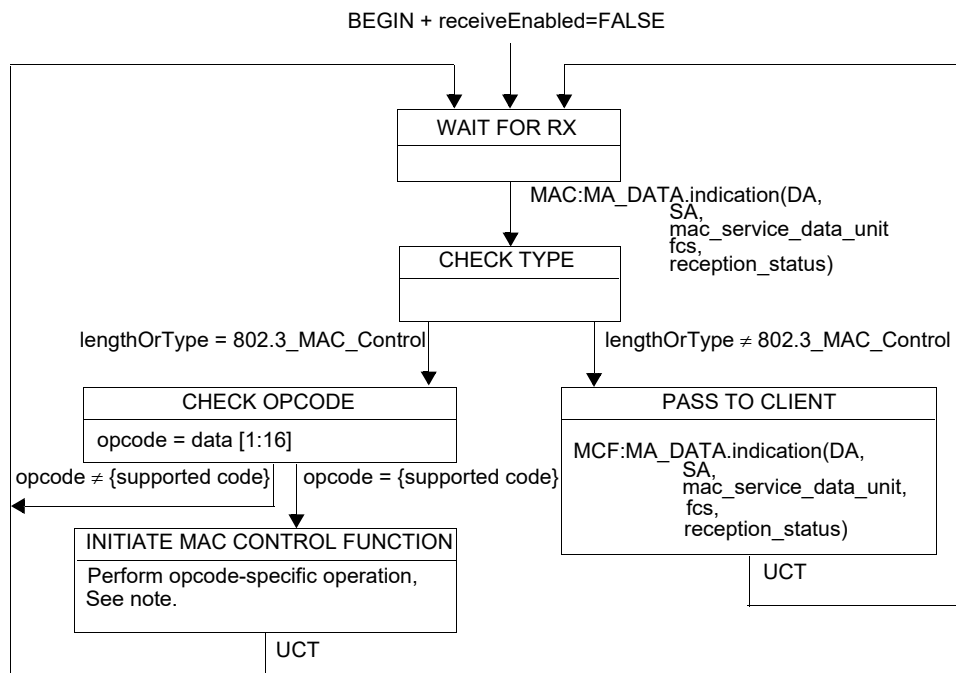
The MAC Control opcode field parsed from the received frame.

31.5.3.3 Messages

MA_DATA.indication

The service primitive used to pass a validly received MAC frame between the MAC and the MAC Control sublayers, or between the MAC Control sublayer and the MAC client. When generated by the MAC sublayer, this message is used by the MAC Control Receive state diagram as the condition for transition between WAIT FOR RX and CHECK TYPE states. While in the PASS TO CLIENT state, the MAC Control Receive state diagram generates this message to the MAC client.

31.5.3.4 Opcode-independent MAC Control Receive state diagram



Instances of MAC data service interface:
MAC=interface to subordinate sublayer
MCF=interface to MAC client

NOTE—The opcode-specific operation (see Annex 31A) is launched as a parallel process by the MAC Control sublayer, and not as a synchronous function. Progress of the generic MAC Control Receive state diagram (as shown in this figure) is not implicitly impeded by the launching of the opcode-specific function.

Figure 31–4—Generic MAC Control Receive state diagram

The functions performed in the INITIATE MAC CONTROL FUNCTION state are opcode-specific (see Annex 31A).

31.6 Compatibility requirements

An instantiation of the MAC Control sublayer is not required to implement all valid control functions specified in Annex 31A.

31.7 MAC Control client behavior

The MAC Control sublayer uses the services of the underlying connectionless-mode MAC sublayer to exchange both Data and Control frames. The MAC Control sublayer does not provide any mechanism for recovery from lost, discarded, damaged, or delayed frames. It is the responsibility of the MAC Control client to implement mechanisms for dealing with lost, discarded, damaged, and delayed frames, if necessary.

Since implementation of the MAC Control sublayer is optional, a MAC Control client cannot assume the existence of a MAC Control sublayer entity in a peer DTE.

31.8 Protocol implementation conformance statement (PICS) proforma for Clause 31, MAC Control⁶⁴

31.8.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 31, the optional MAC Control sublayer, shall complete the following PICS proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

31.8.2 Identification

31.8.2.1 Implementation identification

Supplier	
Contact point for inquiries about the PICS	
Implementation Name(s) and Version(s)	
Other information necessary for full identification— e.g., name(s) and version(s) for machines and/or operating systems; System Name(s)	
NOTE 1—Only the first three items are required for all implementations, other information may be completed as appropriate in meeting the requirements for the identification. NOTE 2—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

31.8.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 31, MAC Control
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No ☐ Yes ☐ (See Clause 21; The answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	

Date of Statement	
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⁶⁴ *Copyright release for PICS proformas:* Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

31.8.3 PICS proforma for MAC Control frames

31.8.3.1 Support by interlayer interfaces

Item	Feature	Subclause	Value/Comment	Status	Support
SI1	Support for MAC service primitives, MA_-CONTROL.request and MA_CONTROL.indication	31.3	Required	M	Yes []

31.8.3.2 MAC Control frame format

Item	Feature	Subclause	Value/Comment	Status	Support
FF1	Length/Type field	31.4.1.3	2-octet field containing 88-08	M	Yes []
FF2	MAC Control opcode	31.4.1.4	2-octet operation code	M	Yes []
FF3	Number of opcodes	31.4.1.4	1	M	Yes []
FF4	MAC Control parameters	31.4.1.5	MAC Control Parameter field as described	M	Yes []

31.8.3.3 Opcode-independent MAC Control sublayer operation

Item	Feature	Subclause	Value/Comment	Status	Support
SD1	Generic MAC Control receive state diagram	31.5.3	Meets requirements of Figure 31-4	M	Yes []
SD2	MAC Control frame action	31.3	Sink MAC Control frames	M	Yes []

31.8.3.4 Control opcode assignments

Item	Feature	Subclause	Value/Comment	Status	Support
COA1	Opcode values and interpretations	31A	Reserved opcodes not used	M	Yes []